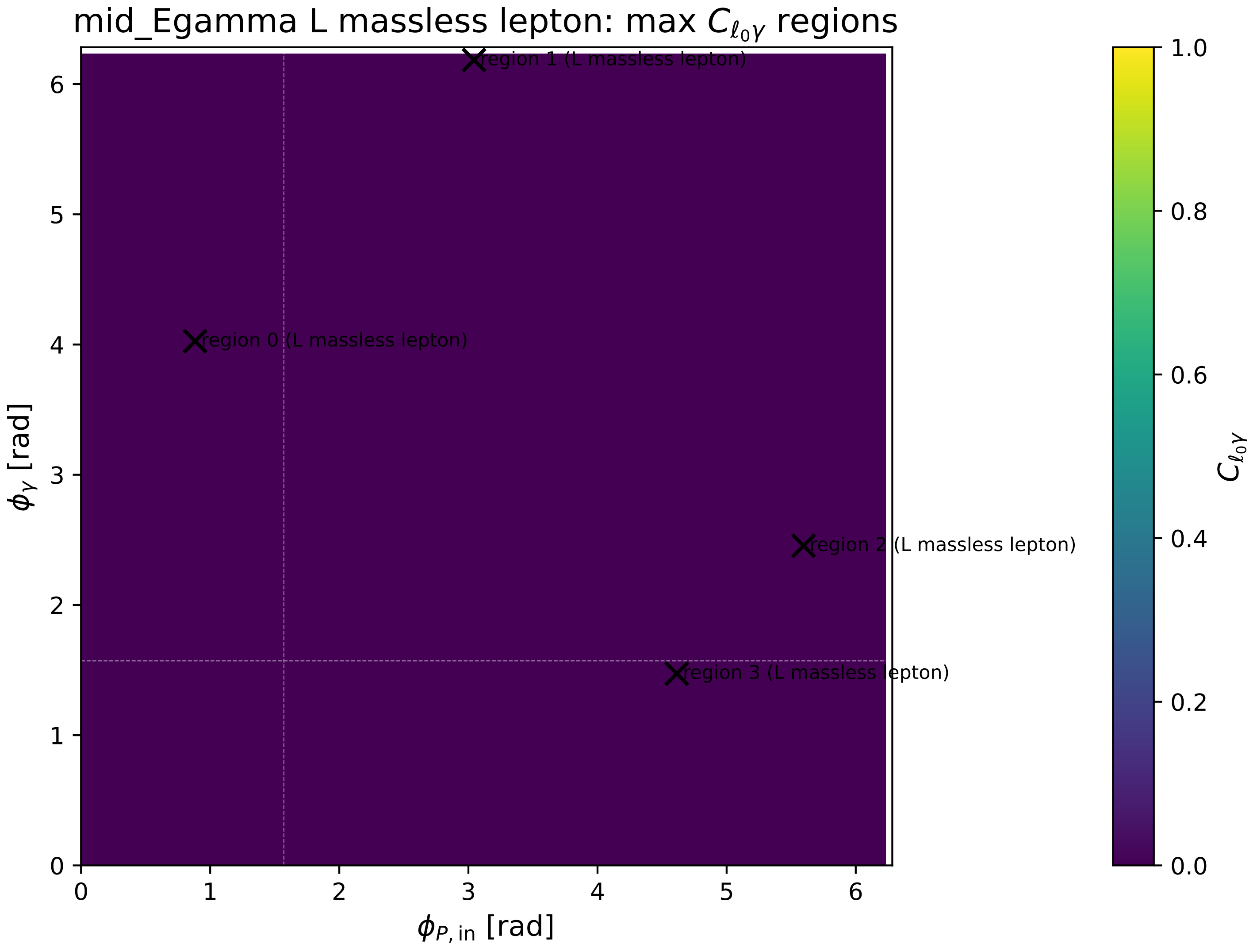
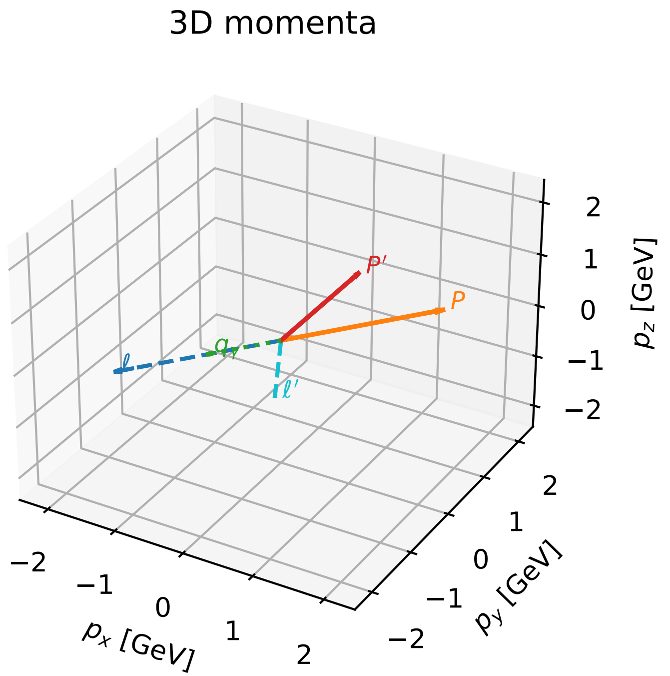

mid_Egamma L massless lepton: max $C_{\ell_0\gamma}$ regions
region 1 (L massless lepton)
region 0 (L massless lepton)
region 2 (L massless lepton)
region 3 (L massless lepton)
$\phi_\gamma$ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$C_{\ell_0\gamma}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

# L massless lepton characteristic max $C_{\ell_0\gamma}$ configuration

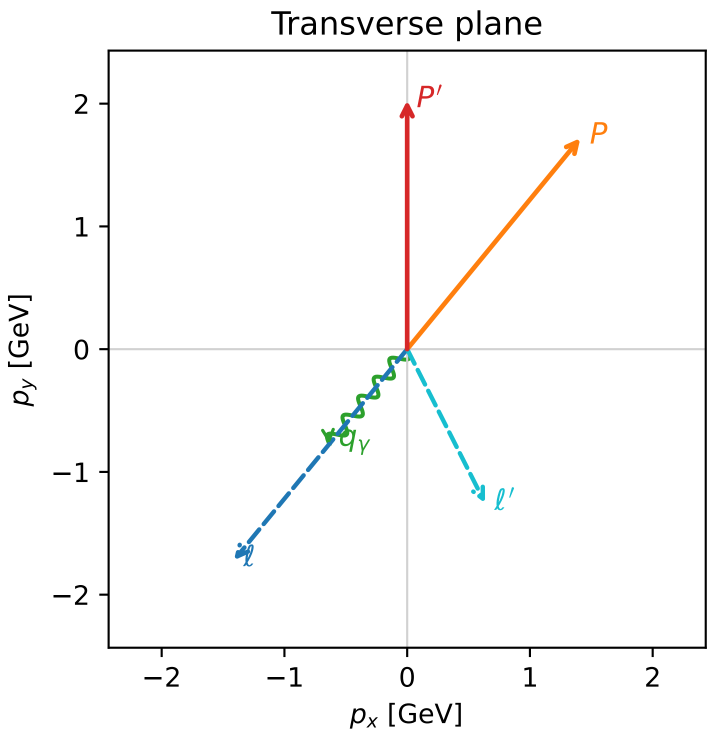


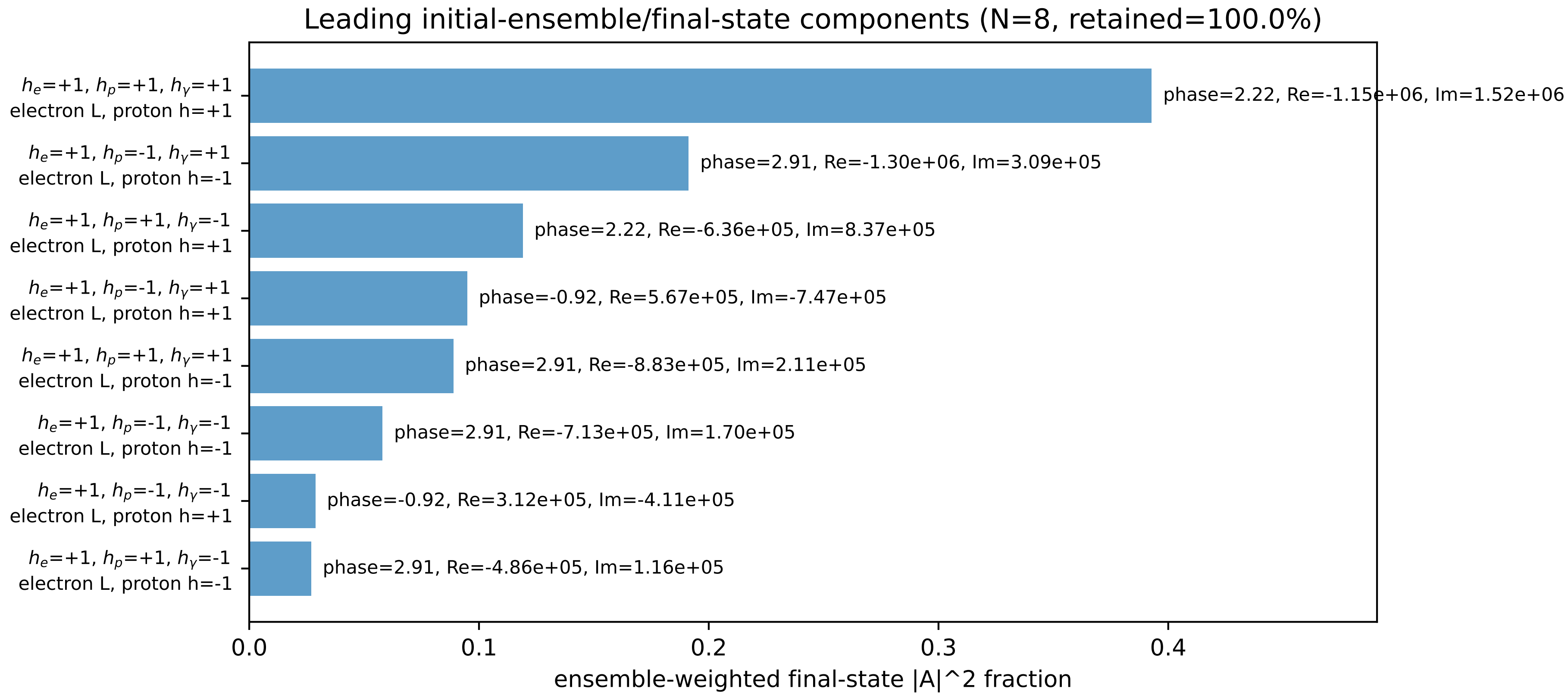

```
c_massless_gamma_mid_egamma_l_lepton_region_0  C_{l_0\gamma}=1.26296e-10
region: mid_Egamma_L_lepton_region_0  final-pair delta_xy=1.15564 rad
outgoing-state purity: 0.545774
kinematic point: high_s_high_theta_in_mid_Egamma
```

incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_{\ell_0}, h_p)$

$s$=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.02682
$\theta_{\rm in}$=1.57079, $\phi_\ell$=4.02517, $\phi_P$=0.883573
$E_\gamma$=1, $\phi_\gamma$=4.02517
$Q^2$=3.72996, $x_B$=0.329206, $t$=-2.05314, $W^2$=8.48005, $y$=0.549049
$\theta(\ell',\gamma)$=66.2131 deg

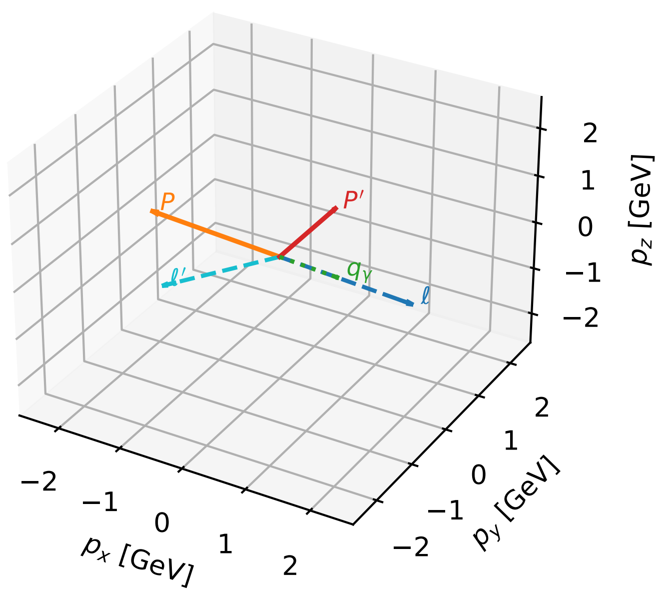
four-momenta [$E$, $p_x$, $p_y$, $p_z$] GeV:
$\ell$ $E$= 2.2244 $p_x$= -1.4112 $p_y$= -1.7195 $p_z$= -0.0000
$P$ $E$= 2.4141 $p_x$= 1.4112 $p_y$= 1.7195 $p_z$= 0.0000
$\ell'$ $E$= 1.4052 $p_x$= 0.6344 $p_y$= -1.2538 $p_z$= 0.0000
$P'$ $E$= 2.2333 $p_x$= 0.0000 $p_y$= 2.0268 $p_z$= 0.0000
$q_\gamma$ $E$= 1.0000 $p_x$= -0.6344 $p_y$= -0.7730 $p_z$= 0.0000



# L massless lepton characteristic max $C_{\ell_0\gamma}$ configuration

3D momenta



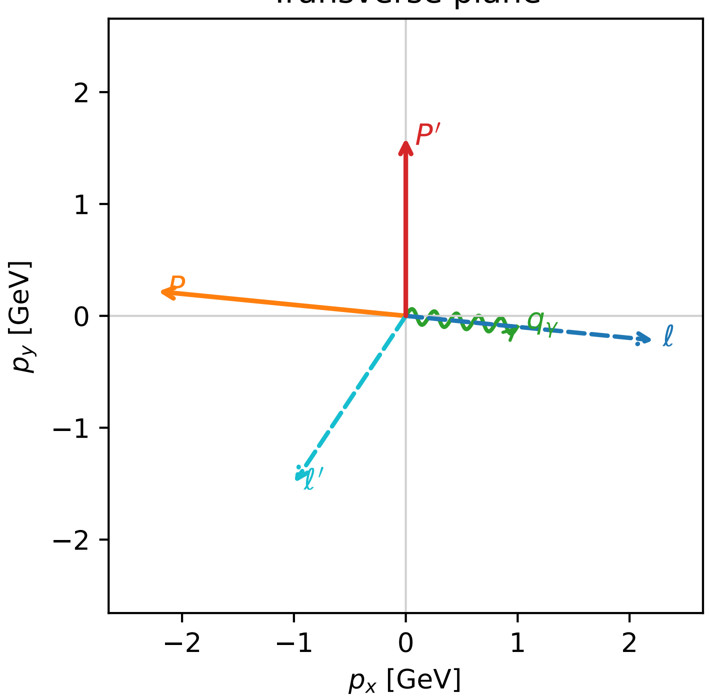

```
c_massless_gamma_mid_egamma_l_lepton_region_1  C_{l_0\gamma}=6.19883e-11
region: mid_Egamma_L_lepton_region_1  final-pair delta_xy=2.06105 rad
outgoing-state purity: 0.814149
kinematic point: high_s_high_theta_in_mid_Egamma
```

incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_{\ell_0}, h_p)$

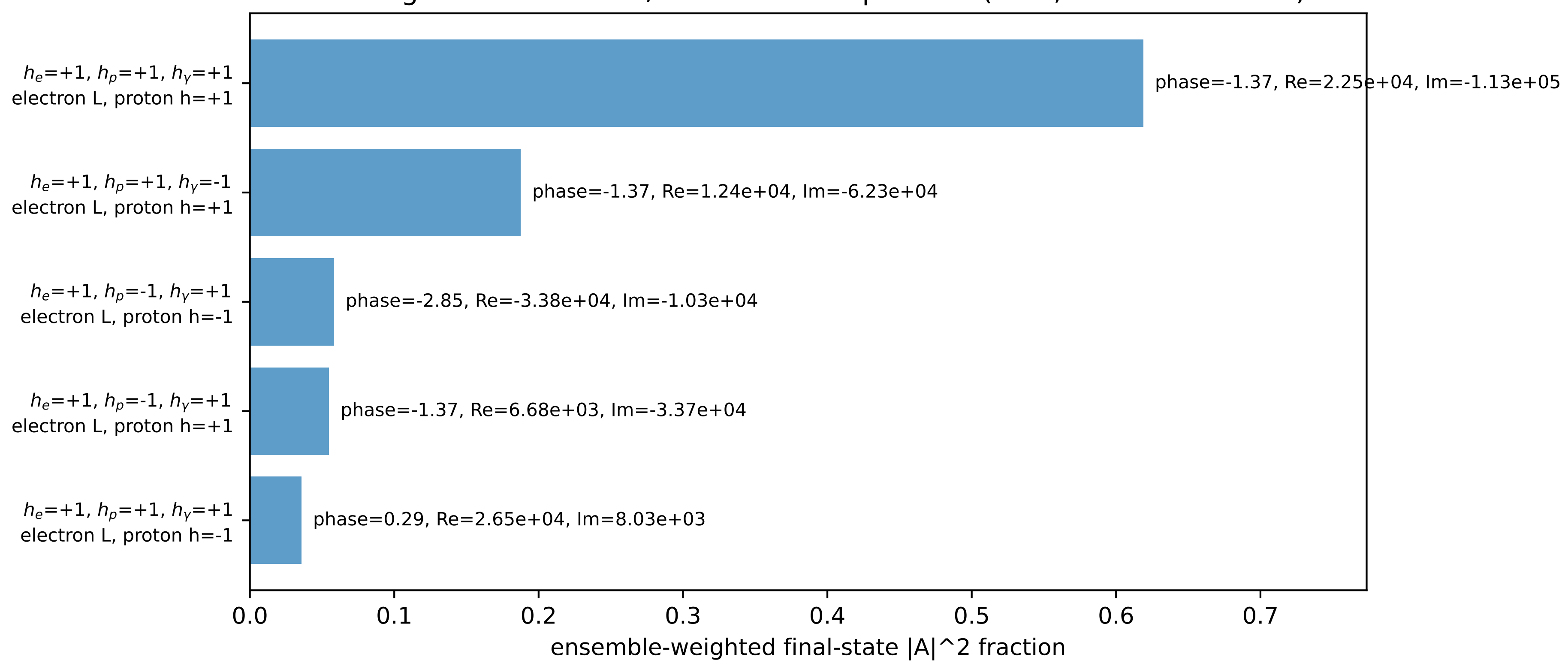
$s$=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.58942
$\theta_{\rm in}$=1.57079, $\phi_\ell$=6.18501, $\phi_P$=3.04342
$E_\gamma$=1, $\phi_\gamma$=6.18501
$Q^2$=11.7323, $x_B$=0.745618, $t$=-6.45798, $W^2$=4.88255, $y$=0.762503
$\theta(\ell', \gamma)$=118.089 deg

four-momenta [$E$, $p_x$, $p_y$, $p_z$] GeV:
$\ell$ $E$= 2.2244 $p_x$= 2.2137 $p_y$= -0.2180 $p_z$= -0.0000
$P$ $E$= 2.4141 $p_x$= -2.2137 $p_y$= 0.2180 $p_z$= 0.0000
$\ell'$ $E$= 1.7930 $p_x$= -0.9952 $p_y$= -1.4914 $p_z$= 0.0000
$P'$ $E$= 1.8456 $p_x$= 0.0000 $p_y$= 1.5894 $p_z$= 0.0000
$q_\gamma$ $E$= 1.0000 $p_x$= 0.9952 $p_y$= -0.0980 $p_z$= 0.0000

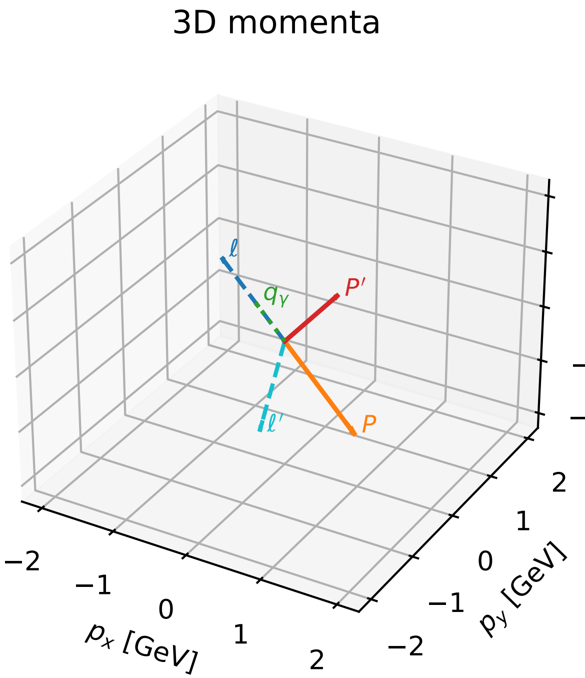
Transverse plane




Leading initial-ensemble/final-state components (N=5, retained=95.5%)



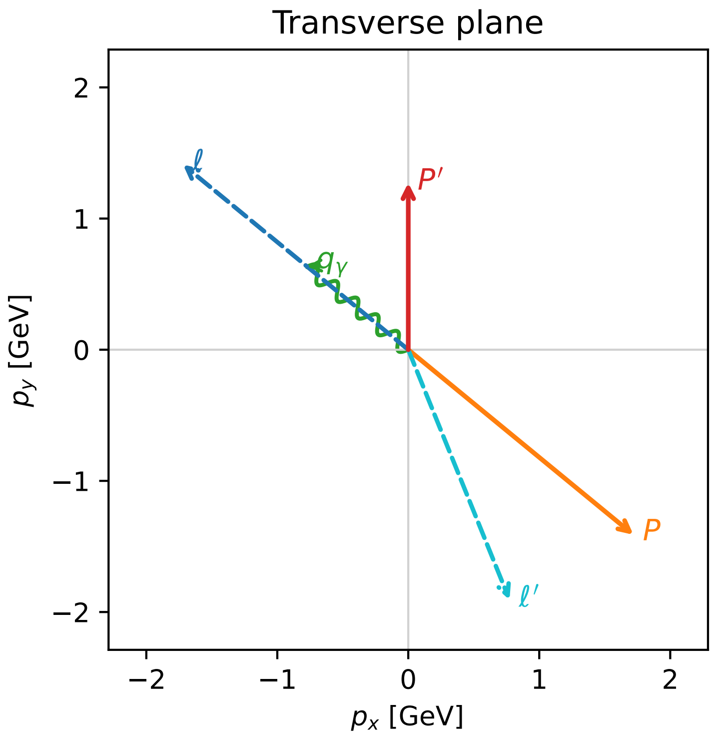
# L massless lepton characteristic max $C_{\ell_0\gamma}$ configuration

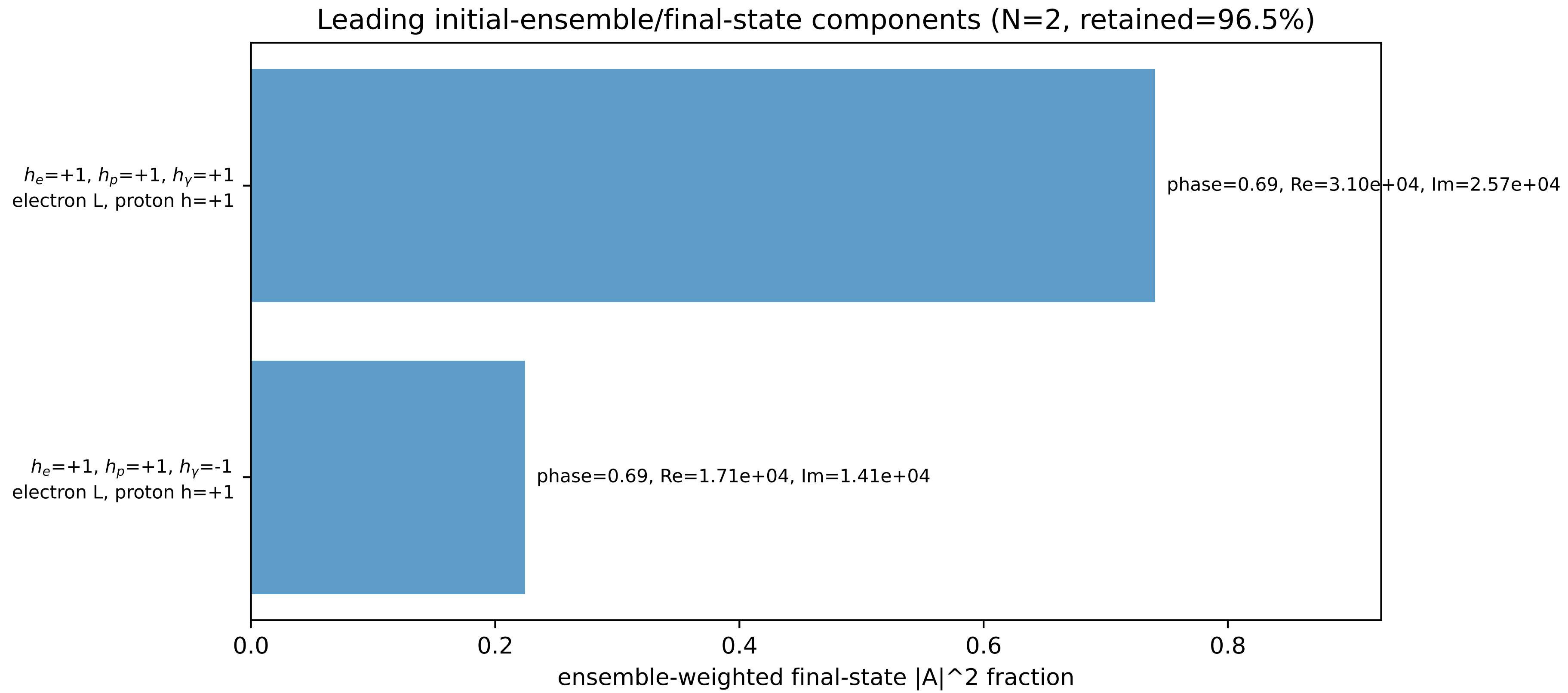



c_massless_gamma_mid_egamma_l_lepton_region_2  C_{l_0\gamma}=3.71701e-11
region: mid_Egamma_L_lepton_region_2  final-pair delta_xy=2.64315 rad
outgoing-state purity: 0.984367
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_{\ell_0}, h_p)$

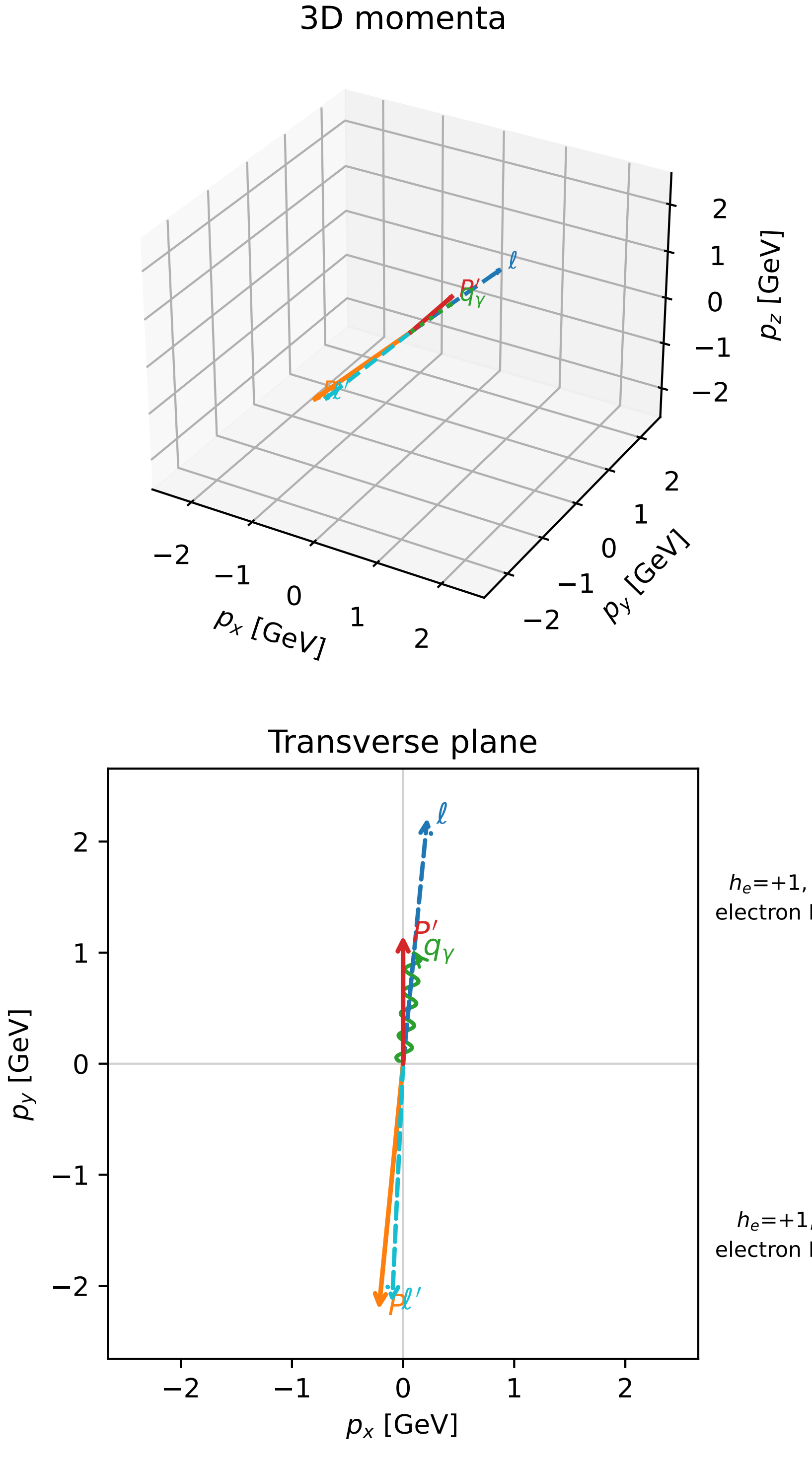
$s$=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.27253
$\theta_{\rm in}$=1.57079, $\phi_\ell$=2.45437, $\phi_P$=5.59596
$E_\gamma$=1, $\phi_\gamma$=2.45437
$Q^2$=17.1944, $x_B$=0.917446, $t$=-9.46456, $W^2$=2.42703, $y$=0.908199
$\theta(\ell', \gamma)$=151.441 deg

four-momenta [$E$, $p_x$, $p_y$, $p_z$] GeV:
$\ell$ $E$= 2.2244 $p_x$= -1.7195 $p_y$= 1.4112 $p_z$= -0.0000
$P$ $E$= 2.4141 $p_x$= 1.7195 $p_y$= -1.4112 $p_z$= 0.0000
$\ell'$ $E$= 2.0576 $p_x$= 0.7730 $p_y$= -1.9069 $p_z$= 0.0000
$P'$ $E$= 1.5809 $p_x$= 0.0000 $p_y$= 1.2725 $p_z$= 0.0000
$q_\gamma$ $E$= 1.0000 $p_x$= -0.7730 $p_y$= 0.6344 $p_z$= 0.0000



# L massless lepton characteristic max $C_{\ell_0\gamma}$ configuration




c_massless_gamma_mid_egamma_l_lepton_region_3  C_{l_0\gamma}=3.30334e-11
region: mid_Egamma_L_lepton_region_3  final-pair delta_xy=3.08899 rad
outgoing-state purity: 0.999998
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_{\ell_0}, h_p)$

$s$=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.15398
$\theta_{\text{in}}$=1.57079, $\phi_\ell$=1.47262, $\phi_P$=4.61421
$E_\gamma$=1, $\phi_\gamma$=1.47262
$Q^2$=19.1292, $x_B$=0.9658, $t$=-10.5296, $W^2$=1.55723, $y$=0.959808
$\theta(\ell', \gamma)$=176.986 deg

four-momenta [$E$, $p_x$, $p_y$, $p_z$] GeV:
$\ell$ $E$= 2.2244 $p_x$= 0.2180 $p_y$= 2.2137 $p_z$= -0.0000
$P$ $E$= 2.4141 $p_x$= -0.2180 $p_y$= -2.2137 $p_z$= 0.0000
$\ell'$ $E$= 2.1514 $p_x$= -0.0980 $p_y$= -2.1492 $p_z$= 0.0000
$P'$ $E$= 1.4871 $p_x$= 0.0000 $p_y$= 1.1540 $p_z$= 0.0000
$q_\gamma$ $E$= 1.0000 $p_x$= 0.0980 $p_y$= 0.9952 $p_z$= 0.0000